

```
from tabulate import tabulate
```

```
def fractional_knapsack(items, capacity):  
    # Sort items by value/weight ratio (descending)  
    items.sort(key=lambda x: x['profit'] / x['weight'], reverse=True)  
  
    total_value = 0.0  
    knapsack_items = []  
    remaining_capacity = capacity  
  
    for item in items:  
        if remaining_capacity == 0:  
            break  
  
        if item['weight'] <= remaining_capacity:  
            # Take full item  
            total_value += item['profit']  
            knapsack_items.append({  
                'id': item['id'],  
                'total_profit': item['profit'],  
                'total_weight': item['weight'],  
                'fraction': 1.0,  
                'weight_taken': item['weight'],  
                'profit_gained': item['profit']  
            })  
            remaining_capacity -= item['weight']  
        else:  
            # Take fractional part  
            fraction = remaining_capacity / item['weight']  
            profit_gained = item['profit'] * fraction  
            weight_taken = remaining_capacity  
            total_value += profit_gained  
            knapsack_items.append({
```

```

        'id': item['id'],
        'total_profit': item['profit'],
        'total_weight': item['weight'],
        'fraction': fraction,
        'weight_taken': weight_taken,
        'profit_gained': profit_gained
    })
    remaining_capacity = 0

```

```

return total_value, knapsack_items

```

```

# ----- USER INPUT SECTION -----

```

```

print("\n--- Fractional Knapsack Problem ---")

```

```

n = int(input("Enter number of items: "))

```

```

items = []

```

```

for i in range(1, n + 1):

```

```

    profit = float(input(f"\nEnter profit for item {i}: "))

```

```

    while True:

```

```

        weight = float(input(f"Enter weight for item {i} (must be > 0): "))

```

```

        if weight > 0:

```

```

            break

```

```

        print("Weight must be greater than zero. Please enter again.")

```

```

    items.append({'id': i, 'profit': profit, 'weight': weight})

```

```

capacity = float(input("\nEnter knapsack capacity: "))

```

```

# ----- DISPLAY INPUT TABLE -----

```

```

print("\nInput Items Table:")

```

```

input_table = [

```

```

[item['id'], item['profit'], item['weight'], round(item['profit'] / item['weight'], 2)]
    for item in items
]

print(tabulate(input_table, headers=["Item ID", "Profit", "Weight", "Profit/Weight"],
tablefmt="fancy_grid"))

# ----- SOLVE THE PROBLEM -----

max_value, taken_items = fractional_knapsack(items, capacity)

# ----- DISPLAY OUTPUT TABLE -----

print("\nItems Taken in the Knapsack (with calculations):")

output_table = [
    [
        item['id'],
        item['total_profit'],
        item['total_weight'],
        f"{item['fraction']:.2f}",
        f"{item['weight_taken']:.2f}",
        f"{item['profit_gained']:.2f}"
    ]
    for item in taken_items
]

print(tabulate(output_table, headers=[
    "Item ID", "Total Profit", "Total Weight", "Fraction Taken",
    "Weight Taken", "Profit Gained"
], tablefmt="fancy_grid"))

print(f"\nMaximum Profit: {max_value:.2f}")

```

OUTPUT:

--- Fractional Knapsack Problem ---

Enter number of items: 4

Enter profit for item 1: 10

Enter weight for item 1 (must be > 0): 7

Enter profit for item 2: 12

Enter weight for item 2 (must be > 0): 7

Enter profit for item 3: 15

Enter weight for item 3 (must be > 0): 4

Enter profit for item 4: 20

Enter weight for item 4 (must be > 0): 13

Enter knapsack capacity: 30

Input Items Table:

Item ID	Profit	Weight	Profit/Weight
1	10	7	1.43
2	12	7	1.71
3	15	4	3.75
4	20	13	1.54

Item ID	Total Profit	Total Weight	Fraction Taken	Weight Taken	Profit Gained
3	15	4	1	4	15
2	12	7	1	7	12
4	20	13	1	13	20
1	10	7	0.86	6	8.57